Machine language as somniloquy: How the neuroscience of sleep illuminates generative AI

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Abstract. Can machines mean what they say? Generally not, because current AI systems lack the cognitive capacities that allow humans to recover meaning from and infuse meaning into language. However, from a cognitive standpoint, human and machine language are not entirely dissimilar. Specifically, while at present there are hardly circumstances in which machines function as humans typically do vis-à-vis meaningful language use, there is at least a circumstance in which humans appear to function as current generative language AI: during *somniloquy* or 'sleep talk'. I first show that somniloquy and machine language unfold in similar restricted cognitive conditions. I then use this observation to suggest that AI's linguistic output should be given a similar 'semantic status' to human somniloquy, and I consider some implications of this idea.

Keywords: Language; Meaning; AI; Neuroscience; Sleep.

1. Can machines mean what they say?

We engage with chatbots and virtual agents as if they meant what they say, or as if there was someone on the other side who reads or listens and understands us, who forms intentions about what to say in reply, and who does so trying to be relevant and helpful. But the problem is, there really is nobody there. A language model, the kind of neural network-based architecture that underlies current conversational AI systems, is nothing like a person to which we could credit those mental states.

And perhaps we do not. Perhaps most of us are not victims of the kind of "powerful delusional thinking" that MIT computer scientist Joseph Weizenbaum ascribed to users of his chatbot Eliza (Weizenbaum, 1967, p. 7). As philosopher Daniel Dennett (1987) observed, we tend to take an *intentional stance* towards all sorts of systems, including AIs: we try to explain their behaviour in terms of beliefs, intentions, and desires. We could go even further and describe what chatbots and virtual agents do as 'role playing', or impersonating the characters that appear to be in conversation with us, as Murray Shanahan, Kyle McDonell, and Laria Reynolds (2023) recently argued. But if the chatbot is impersonating a character, then so are we: we may not believe we are *actually* engaging in a meaningful exchange with another agent; we just act *as if* we were, and we describe the AI's and our own behaviour accordingly.

Machines cannot mean what they say, or anything at all, for the simple reason that they lack the cognitive capacities that, in humans, make meaning possible, such as the ability to form communicative intentions, to refer to specific entities or issues, and say specific things about them. Current AIs cannot recover meaning from what we say, nor can they infuse meaning into what they say to us in response. Still, this semantic impasse is overcome by an implicit pretend play at the basis of which lies an act of taking a *semantic stance*: we treat patterns that look like language, in the broadest sense, *as if* they had meaning, not just on their own, but in virtue of some agent that uses such patterns to communicate through them. Conversation with an AI is not quite a 'powerful delusion', rather a game of make-believe in which many of us engage wittingly: meaning attributions are the moves we make in this game.

2. Somniloquy and the neuroscience of sleep

If AIs cannot intend what they say, or anything at all, and it is rather us who do all the semantic heavy-lifting as we chat with them, it would seem that is because they are so completely unlike us cognitively, that whatever it is they do with language, it has no human analogue or counterpart. There is some truth in this type of answer, but it is not entirely satisfactory. On the one hand, it sets up a hard divide between machines and humans that may be useful in reminding us that we should not treat them as one of us, and that we should not regard humans as machines of that type. On the other hand, the same distinction risks obfuscating our understanding of the variety of conditions in which language is used. Ultimately, all such conditions, no matter how alien they may seem, share something characteristically human. Here, I attempt to show that even the peculiar ways in which current AIs do language are possible human ways of doing language. In very specific circumstances, we may be machine-like after all: in that sense, machines would be human-like. I will leverage the observation that, from a cognitive angle, human and machine language are not entirely disjoint. At present, there are hardly any circumstances in which machines function as we normally do vis-à-vis meaningful language use, but there is at least one specific circumstance in which humans seem to operate as generative language AI systems: during *somniloquy*, or 'sleep talk'.

To motivate this idea, we have to take a detour through the neuroscience of human sleep. Every 90-100 minutes, the brain cycles through each of 4 non-REM (NREM; non-rapid-eye-movement) sleep stages and one period of REM sleep. The duration of the latter increases through the night, whereas the depth and duration of NREM sleep decrease. In NREM stages 1 and 2, muscle activity, blood pressure, and body temperature gradually decrease, and conscious awareness of the body and external environment dissolve. In NREM stages 3-4, often referred to as 'deep sleep', brain activity is most different from wakefulness, with a prevalence of slow waves in the electroencephalogram (EEG). Cognitive activity is also altered: dreaming can occur in these NREM stages, too, but is usually less vivid, complex, and coherent than in

REM sleep. In REM sleep, bodily and brain physiology appear to regain several of the characteristics of wakefulness, such as less regular breathing, higher heart rate, and more complex EEG: that is why REM sleep is also called ‘paradoxical sleep’. In this stage, dreams are often more bizarre and memorable, with distinct scenes and narrative structures. Yet, remembering one’s dreams remains unlikely, because the neurotransmitter systems that support short-term memory during wakefulness are inactivated during sleep. Sudden awakening restores the availability of the relevant molecules to short-term memory structures in the brain, making it more likely that one will remember the dreams one was having immediately before waking up.

Dreaming is not the only kind of cognitive activity that may occur during NREM or REM sleep. Somniloquy or sleep talk is equally common, if not as significant at the personal level. Somniloquy is no longer included in NREM parasomnias, according to the ICSD-3 (International Classification of Sleep Disorders), and is listed under “isolated symptoms and normal variants”. Around 60% of people report talking in their sleep at some point in their lives, which is approximately the same prevalence as nightmares. Sleep talking can occur during both NREM and REM sleep.

Typical vocalisations during somniloquy range from 1 to 5 words, and each episode can last for up to 30 seconds. Productions can include both meaningless sequences of words and sounds as well as fully articulated, grammatically correct phrases and sentences. Recent studies show that most intelligible utterances in somniloquy are grammatically correct, and that sleep talking might even respect turn-taking with a partner who may also report or record the conversation. This indicates that certain linguistic (syntactic, semantic, and pragmatic) processes are largely automatic and do not depend on consciousness or cognitive control. Most of the clauses produced during somniloquy contain a verb, most frequently in the present tense (more than 80% of all verbs). About 90% of clauses are addressed to somebody other than the speaker, with frequent use of second person pronouns.

‘The Further Somniloquies of Dion McGregor’, freely available on the internet, are a famous and extensive collection of examples of sleep talk, which may however be not entirely representative of typical somniloquy. Two examples from the scientific literature are given below (Arkin, 1982). The first is reported as a dialogue between a sleep talker ('he') and his awake partner ('she'):

(1) He: Boy—do I have an idea!
She (after a brief pause): What idea, dear—tell me about it.
He: The workers in the east, and the workers in the west—they don't know about it.
She: Tell me, honey—what is it?
He (after a pregnant pause): Why should I tell you?

(2) "Peacock tree?! There's a tree full of peacocks and swallows and other colorations, aabsolutely, aaaabsolutely g-o-o-or-geous – get my camera, Larry, and don't forget to turn the film or I'll have fits when it gets back – come on! God! – you should see it." [This was uttered with an air of brisk alert enthusiasm.]

Somniloquy during NREM sleep is most interesting for our purposes here, because during NREM sleep, people are, in important cognitive respects, not unlike current disembodied AI systems: sensation and perception are dulled or absent, attention is lost, thought may be logical but is typically perseverative, orientation (awareness of time, place, person) is unstable or absent, and emotions are weak and indistinct. In REM sleep, some of these functions are regained, but are endogenously driven: this is another scenario in which humans are radically unlike current machines. In REM sleep, movement cannot normally occur, preventing the body from acting out dream contents. In contrast, in NREM sleep movement can occur — this is the case of sleepwalking as well as of sleep talking. NREM somniloquy is not necessarily the overt counterpart of utterances in dreams (dream speech), even though overt sleep talking during REM sleep can also occur, if the brain stem fails to paralyze relevant muscles. Sleep talkers only rarely recognize reports of what they have been saying in sleep as matching the contents of the dreams (if any) that they were having just before waking up. But the association between somniloquy and NREM sleep seems fairly clear. Sleep talking often begins as a partial, confusional arousal out of stage 3 slow wave sleep, and then may continue with some characteristics of REM sleep or wakefulness, including coherent cortical or motor activity, such as overt speech. Further, when the frequency and duration of NREM sleep increase, for example in childhood, so does the likelihood of overt motor activity, in particular sleepwalking and sleep talking.

3. Is machine language like somniloquy?

The hypothesis here is that *somniloquy and machine language unfold in cognitive conditions with similar restrictions*, even though those restrictions are in place for very different reasons. Both generative AI systems and people during NREM sleep are unconscious, they have no sense of a body or a self, no external perceptions, no sense of the environment, no orientation (no sense of time, space, or other people), no volition or motivation, and hence no capacity to form communicative intentions and act upon them. Language is not understood, but can be generated even in such conditions of 'minimal cognition': machines and sleeping people can talk, and that shows that the cognitive regime of wakefulness is not necessary for grammatically correct and (for an interpreter) meaningful language.

Machine language, the output of current large language models and the AI systems that are based on them, is like somniloquy in a number of significant ways. From a linguistic point of view, the two could not be more different. AI texts can be as long and sophisticated as one desires, can be produced in several different registers and languages, may contain code, formulae, or diagrams, and are generally well aligned with users' expectations. Reinforcement Learning from Human Feedback (RLHF), for example, which trains a chatbot on appropriate prompt-response pairs, almost guarantees that profanities, of the kind that can occur in somniloquy, will never be produced. But there are also several similarities: machine language tends to prefer the present tense; it addresses the user, an inaccessible person, as 'you'; it respects grammar and turn-taking; and can be disturbingly disconnected from reality, logic, or common sense, as evidenced by so-called 'hallucinations'. These are all features of sleep talk, too. And if we consider the earlier, rawer GPT- or BERT-type models, that have not been fine-tuned or aligned with human preferences through RLHF or other techniques, the similarities become even more apparent.

But more important than these similarities is the underlying cognitive resemblance between a sleeping brain and a computer running a language model. In both cases, albeit for different reasons, the system is 'on' and generating language, but there is no consciousness, no sense of self, body, or environment, of time, space, or others, no intention, volition, motivation, or control. Still, the cognitive wheels of language are spinning, and what they produce reflects the rules and patterns the system has internalised. Thus, both a sleeping person and a chatbot may use words such as 'I', 'you', 'here', 'now', 'in pain', 'away' etc., that refer to those aspects of self and world that are dimmed or absent for them, but their talk would not make sense. For what would 'here', as generated by a language AI, mean, when the model has or can have no notion of where it is in space? The same would apply to a sleeping person who says 'here': are they referring to their actual sleeping location, to their location in a dream, none of these, or no location at all? For an AI and a human in NREM sleep, there is no first person with a body and mind, so no meaning for those expressions that require that. There is no volition or intention: an AI or a sleeping person could not intend to be ironic, nor happen to be so in spite of their intentions. There is no environment, so no way to refer to people, things, and events around them. Certain types of talk stop making sense when uttered by machines or sleeping people, and for similar reasons: they both lack, temporarily or permanently, the infrastructure of meaningful language use, that is, cognitive and brain systems for consciousness, intentionality, volition, motivation, sense of self, body, and surroundings.

4. Waking up to truth and other norms

Of course, not everything machines and sleep talkers say is nonsense. Shakespeare explored this idea as he envisaged Lady Macbeth walking in her sleep, rubbing her hands as if to wash them, even confessing to her participation in the assassinations of Lady Macduff, King Duncan, and Banquo:

> "Here's the smell of the blood still. All the perfumes of Arabia will not sweeten this little hand. Oh! Oh! Oh!"

> "Wash your hands; put on your nightgown; look not so pale. I tell you yet again, Banquo's buried; he cannot come out on's grave. To bed, to bed! There's knocking at the gate. Come, come, come, come, give me your hand. What's done cannot be undone. To bed, to bed, to bed."

Lady Macbeth's speech is not only coherent and meaningful, but also true — if not quite truthful, because it would seem that truthfulness requires a capacity to form communicative intentions, such as the intentions to inform or deceive. But indeed, intentions are not necessary for utterances to be evaluated for meaning, truth, and other norms — relevance, plausibility, politeness etc. If someone in their sleep says 'Picasso was an impressionist', they would utter something false about Picasso, and so would a machine generating the same sentence. The fact that what sleep talkers and AIs say or write is generated in restricted cognitive conditions does not entail a complete suspension of semantic norms. This raises some difficult questions: e.g., in law, whether statements made in sleep could count as testimony; or in attempts to automate aspects of scientific discovery, when claims made by AI systems, such as 'This analysis contains evidence for methane on Mars', should be taken literally, as statements that may be true or false, or as hallucinations generated on the basis of training data that include a large number of statements on the subject.

Ordinary and scientific human language, too, may be meaningful, and thus true or false, or nonsensical. In somniloquy and AI-generated language, the probability of nonsense is however much higher, precisely because the mechanisms that generate language operate in isolation, disconnected from systems of knowledge, reasoning, intentionality, a sense of self and others, a sense of one's surroundings etc. Unless there are reasons, provided by serious research, to think otherwise, we should treat machine text and speech with the same skepticism we apply to somniloquy, at least for most current AI systems. That is, machine language ought to be given a similar 'semantic status' to somniloquy. If it has any coherent, interpretable meaning, that is so not because that was the machine's communicative intention, but because the mechanism generating language exploits learned statistical patterns, which make a relevant text output somewhat more likely than an irrelevant one, given the user's prompt. Likewise, if it happens to be true or false, that is not because the AI had an intention to inform or deceive. The prospects of entrusting creative processes to an AI system that operates like a sleep talker may be exciting, but something would be amiss if the AI's task was, say, to generate scientific hypotheses, which ought to be, explicitly and by design, aimed at truth or falsehood, corroboration or falsification, higher or lower prior probability etc. If the analogy between machine language and somniloquy holds, it is not clear that it would be rational for us to assign machines tasks of that kind, much as we would never assign them to agents that, temporarily or permanently, lack the requisite cognitive capacities.

In this article, instead of pursuing the idea that human and machine cognition and language are entirely different, I have explored the proposition that there could be overlap between them. The question is not: in what circumstances could machines do language as humans, and break their current limitations? Rather, the question is: in what circumstances do humans do language like current machines, and show similar restrictions? The answer is during NREM sleep. Human consciousness and cognition in NREM sleep offer little support to the brain mechanisms that generate language: during sleep talk, they operate in a 'cognitive vacuum' that resembles in several important respects the way current generative AI works. Machine language is therefore not unlike somniloquy, and for all practical purposes should be treated as such, in the absence of guarantees that would imply a different stance.

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